Highlights

Shared presentation order leaks through subject-independent evaluation of fNIRS workload decoders: a multi-dataset audit with a null-window probe

Pinzhen He

- Eight public EEG/fNIRS datasets present one block order to every participant.
- Leave-one-subject-out testing does not remove information carried by that order.
- A pre-onset null-window probe detects it in three fNIRS datasets (+6 to +7 points).
- Most of one published fNIRS benchmark margin is available before blocks start.
- The effect depends on pipeline: one published model was unaffected.

Shared presentation order leaks through subject-independent evaluation of fNIRS workload decoders: a multi-dataset audit with a null-window probe

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Abstract

Background: Subject-independent evaluation (e.g. leave-one-subject-out, LOSO) is regarded as protection against leakage in brain decoding. Yet several public EEG/fNIRS datasets present the same condition sequence to every participant, so session position, drift and carry-over can predict labels in a held-out subject.

New method: A model-agnostic null-window probe: LOSO decoding of the upcoming condition from a window that ends before the block and any cue, combined with causal or no filtering, label-free offset removal, a drift-injection positive control, a per-subject-order negative control, and two tests needing no pre-onset window (task-to-rest transfer; reversed block pairs).

Results: At least eight public datasets share one order. In three fNIRS datasets the upcoming condition was decodable before onset: Shin 2018 n-back (+6.3 points, causal filter, $n = 26$), Tufts fNIRS2MW (+6.9, causal baseline, $n = 68$), and Chen 2023 Stroop, where decoding fell to chance on temporally reversed pairs ($n = 18$). Conventional preprocessing gave 12–17 points. With per-subject order the probe was at chance; EEG showed no effect. About 80% of a published BenchNIRS margin was available 12 s before onset; the official Tufts DeepConvNet tracked session position as strongly as load; Con4m was unaffected.

Comparison with existing methods: Earlier work showed within-subject temporal confounds and proposed subject-independent splits as the remedy; this remedy fails under shared order.

Conclusions: Holding out a participant does not hold out a shared sequence. Order should be randomised per participant, and decoding studies should report a null-window or reversed-pair control.

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Keywords: fNIRS, EEG, mental workload, data leakage, cross-validation, experimental design, benchmark

1. Introduction

Decoding mental workload from functional near-infrared spectroscopy (fNIRS), alone or together with EEG, is an active field with a growing number of public benchmarks [1, 2, 3]. Reported accuracies on the same data differ widely. Within-subject evaluations on the Shin et al. n-back data often exceed 90%, while careful subject-independent evaluations of the same or similar problems land between 39% and 56% for three classes [4, 5]. That gap is usually attributed to subject-specific signal and to leakage between temporally adjacent training and test samples, and subject-independent evaluation is commonly prescribed as the remedy [6].

Temporal confounds in block designs are well documented. When all trials of a class occur in one contiguous block, a classifier can decode recording time instead of the stimulus [7, 8, 9], and filtering can move information backwards across event onsets [10]. These demonstrations are within-subject. When they were extended across subjects, accuracy fell to chance [7, 8, 9]. Subject-independent splitting has therefore come to be seen as a fix.

That conclusion holds only if the test subject's temporal structure is unrelated to the labels learned from the training subjects. It is not when every participant receives the same condition sequence. Position in the session, the identity of the preceding block, the length of the rest that follows it, and slow drift are then aligned with the label in the same way for everyone. A decoder trained on some subjects can exploit that alignment in a new subject. The mechanism was noted in a methods paragraph of an fMRI study [11]. To our knowledge it has not been quantified, given a diagnostic, or examined in public EEG/fNIRS benchmarks.

Here we (i) survey public EEG/fNIRS datasets for shared presentation order; (ii) introduce a null-window probe and controls; (iii) measure order information in three fNIRS datasets; and (iv) re-run three published decoders to see whether their reported performance depends on it.

2. Materials and methods

2.1. Census of presentation order

We examined public EEG and fNIRS datasets that are used for decoding and are distributed with continuous recordings or event timing. Where

possible, order was read from the released event files (BrainVision markers, NIRx event files, BIDS `events.tsv`, MATLAB marker structures) rather than from the descriptor paper. Where only documentation was available this is stated.

2.2. Datasets analysed

Shin et al. 2018 n-back [1]: 26 participants, simultaneous EEG (28 channels, 1000 Hz) and fNIRS (36 channels, 10.4 Hz); 0-, 2- and 3-back blocks of 40 s preceded by a 2 s instruction showing the task type and followed by 20 s rest; three runs of nine blocks. Word generation (WG) from the same session served as negative control (below).

Tufts fNIRS2MW [2]: 68 eligible participants, frequency-domain fNIRS (two forehead probes, intensity and phase at two wavelengths, 5.2 Hz); 16 n-back blocks (0–3-back, ∼83 s). For the null-window probe we used the raw release (Section 3.2). For benchmark analyses we used the official band-pass-filtered task segments and the official 30 s sliding windows.

Chen et al. 2023 Stroop [3]: 21 participants, EEG (34 channels) and fNIRS (20 channels, 100 Hz); four 32 s blocks of a Chinese colour–word Stroop task (neutral or incongruent), separated by ∼24 s breaks. We analysed raw fNIRS optical density.

PhysioNet mental-fnirs [12]: 12 participants with fNIRS; four ∼134 s blocks in the fixed order 0-, 1-, 0-, 2-back.

2.3. Null-window probe

For each block we extracted a 4 s window that ends before the block and before any cue about it. In Shin, the window ends 2 s (the instruction) plus a gap $g \in \{0, 2, 4\}$ s before the marked onset. In Tufts, it ends at the start of the instruction screen, inside the rest that follows the previous block. Features were the per-channel mean and least-squares slope within the window. For EEG they were log band power (theta, alpha, beta) per channel.

Three preprocessing choices determine whether such a window can contain post-onset or label information through the analysis itself, and each was controlled: (a) *Filtering.* Zero-phase filters can move post-onset activity into the window [10]. We compared zero-phase, causal (minimum-phase) band-pass, causal low-pass only, and no filtering. (b) *Baseline.* Per-block baselines taken just before onset overlap the window, so no per-block baseline was used. Between-subject offsets were removed by subtracting a label-free per-recording constant: either the recording mean, or a strictly causal

mean over data recorded before all probe windows. (c) *Cue.* Windows never overlapped the instruction or cue period.

Classification was LOSO with standardisation and L2-regularised logistic regression; C was chosen by inner subject-grouped 5-fold cross-validation. The unit of analysis is the participant. We used one-sided Wilcoxon signed-rank tests of per-subject accuracy against chance. For Tufts we additionally used a permutation null in which labels were permuted independently within each participant (500 permutations), which corresponds to per-subject randomised order.

2.4. Positive and negative controls

Drift injection (positive control). In Tufts we shuffled each participant's three null-window feature vectors across positions, which destroys any real order information. We then added k times the observed group-mean change per position ($k \in \{0, 0.25, 0.5, 1, 2\}$, 20 repetitions) and labelled positions by the shared order.

Per-subject order (negative control). The Shin WG task was recorded in the same session, with the same participants, montage and preprocessing, but its trial order is randomised per participant (24 distinct sequences among 26 participants, read from the markers). We report the probe for the current label. We also report two quantities that the shared randomisation *rule* supplies without any data: the label differs from the previous one with probability 0.62 for everyone, and each run contains exactly ten trials of each class.

2.5. Tests that do not rely on a pre-onset window

Task–rest transfer (Tufts). Using the raw final four blocks of all 68 participants, we trained on task windows or on rest-end windows and tested on either, giving four combinations (LOSO).

Reversed pairs (Chen). The released event codes and the authors' per-condition preprocessed files show that every participant saw neutral, incongruent, neutral, incongruent (Section 3.1). Neutral–incongruent pairs therefore occur in two temporal directions: forward (positions 1–2 and 3–4, neutral first) and reverse (2–3, incongruent first). A decoder that reads condition should be equally accurate on both. A decoder that has learnt "later means incongruent" should be accurate on forward pairs and fail on the reverse pair.

Decoder output versus session position (Tufts official data). A 0-vs-2-back decoder was applied to all 16 blocks of each held-out participant. Its block-averaged probability of "0-back" was regressed on true load, block

position and whether the previous block was a 3-back. The model included subject fixed effects, with standard errors clustered by subject.

2.6. Re-running published methods

BenchNIRS [4]: package version 1.7.1 with its own Shin n-back loader and the published "generalised" pipeline (IIR band-pass 0.01–0.5 Hz, TDDR, baseline −2 to 0 s, four ROI channels, mean/std/slope over 0–9.9 s). The only change was to shift every block marker 12 s (or 16 s) earlier before epoching. The feature window then lies entirely before the instruction. Accuracy was scored with the published nested 5-fold scheme and with LOSO for per-subject statistics.

Tufts DeepConvNet [2]: the official architecture, official 30 s windows and the official 17 test buckets (48 train / 16 validation / 4 test participants). To fit CPU time, every third training window was used, the learning rate was searched over $\{0.001, 0.01\}$, and training ran for at most 40 epochs with early stopping on validation accuracy.

Con4m [13]: the authors' code on Tufts 0 vs 2-back (one of their 12 splits; 17 test participants). The official task segments are butted together with rest removed. We re-concatenated each participant's 16 blocks in an independent random order, which leaves block content unchanged, and crossed training and test order. Test databases were regenerated with five sampling seeds (fixed order) and five shuffles (shuffled order), with participant groups held constant.

2.7. Software and data availability

All analyses used Python (MNE, scikit-learn, PyTorch). Scripts `20_`–`32_` and `91_`–`92_` in the accompanying repository reproduce every number and figure from the public datasets. Code: `https://github.com/huixun160/shared-order-audit`.

3. Results

3.1. Shared order is common, and sometimes misdescribed

Table 1 lists the datasets examined. At least eight present one sequence to every participant. They include two widely used fNIRS workload benchmarks (Fig. 1). Descriptions of the order were sometimes incomplete or inaccurate. Shin et al. and Tufts describe their orders as counterbalanced; the balancing is a Latin square within participant, and all participants receive the same square. In Tufts, rest after a block lasted 20, 20, 30 and

40 s after 0-, 1-, 2- and 3-back (medians 20.0/20.0/29.6/39.8 s across all participants), so rest length itself encodes the preceding load. Every non-initial 0-back follows a 3-back. For Chen et al., the paper states the order neutral, neutral, incongruent, incongruent. The EEG event codes (1-2-1-2) and the authors' own per-condition preprocessed files, matched back to the raw recordings (17 of 19 checkable participants), show neutral, incongruent, neutral, incongruent.

Table 1: Presentation order in public EEG/fNIRS datasets. "Data" = verified from released event files in this study; "Doc." = from documentation only.

Dataset	Modality	N	Order	Source
Shared by all participants				
Shin 2018 n-back [1]	EEG+fNIRS	26	9-block sequence per run, identical for all	Data
Tufts fNIRS2MW [2]	fNIRS	68	16 blocks `0123 1230 2301 3012`; rest length depends on load	Data
Tufts NIRS-X (audio n-back)	fNIRS	53	same 16-block sequence	Doc.
Chen 2023 Stroop [3]	EEG+fNIRS	21	N-I-N-I (paper text: N-N-I-I)	Data
PhysioNet mental-fnirs [12]	fNIRS	12	0-1-0-2-back; 2-back always last	Data
Rybář 2025 [14]	EEG+fNIRS	12	identical stimulus sequence for participants 2–12	Data
ZuCo 1.0/2.0 [15]	EEG+eye	12/18	same sentences in same order	Doc.
High-altitude Stroop (Sci Data 2024)	EEG+fNIRS	80	one fixed 120-trial sequence	Doc.
Randomised per participant				
Shin 2018 word generation [1]	EEG+fNIRS	26	24 distinct sequences; 10/10 per run; $P(\text{switch}) = 0.62$	Data
COG-BCI [16]	EEG	29	task order pseudorandomised per session	Data

3.2. Data releases

The Tufts raw release contains the first twelve blocks for only 9 of 68 eligible participants. The final four blocks (3-, 0-, 1-, 2-back) are present for all 68. We verified this against the archive listing, so it is not a download

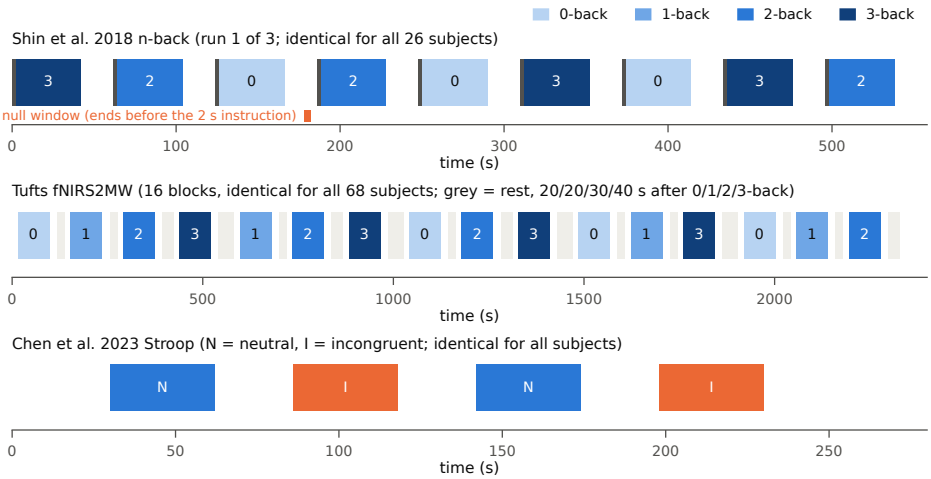


Figure 1: Block sequences shared by all participants in the three datasets analysed. Shin: grey bars mark the 2 s instruction; the orange mark shows a null window, which ends before the instruction. Tufts: grey segments are rest, whose length depends on the preceding load. Chen: the order recovered from the data.

error. The official filtered release keeps only task segments and was filtered zero-phase, so it cannot support a pre-onset probe. Null-window analyses of Tufts therefore use blocks 14–16 (0-, 1-, 2-back) of the raw release, each preceded by rest and an instruction. In these blocks, label and position are aliased, which makes them a strong case of shared order.

3.3. The upcoming condition is decodable before onset in fNIRS

In Shin, fNIRS null windows ending 2 s before the instruction decoded the upcoming load above chance under every filter (Fig. 2A, Table 2): causal band-pass 39.6% ($p = 0.003$), causal low-pass only 39.9% ($p = 0.004$), and no filter 38.8% ($p = 0.004$). Zero-phase band-pass inflated this to 47.2%. EEG from the same windows was at chance (34.3%, $p = 0.24$). A previously used window that ended at onset had suggested a small EEG effect, but it overlapped the instruction.

In Tufts, null windows at the end of rest decoded the upcoming load at 40.2% with a strictly causal baseline ($p = 0.019$) and at 50.0% with the recording-mean baseline ($p = 5 \times 10^{-5}$; within-subject permutation $p = 0.002$) (Fig. 2B). Without any offset removal it was at chance (31.9%). A per-subject constant cannot carry label information, so offset removal reveals, rather than creates, information that sits in the relative level of

the windows. The confusion matrix was ordinal: 0- and 2-back were best separated and 1-back fell in between.

Applying Holm correction to the most conservative test of each dataset (Shin causal, Tufts causal baseline, Chen reversed pairs, BenchNIRS shifted window), all four remained below 0.05.

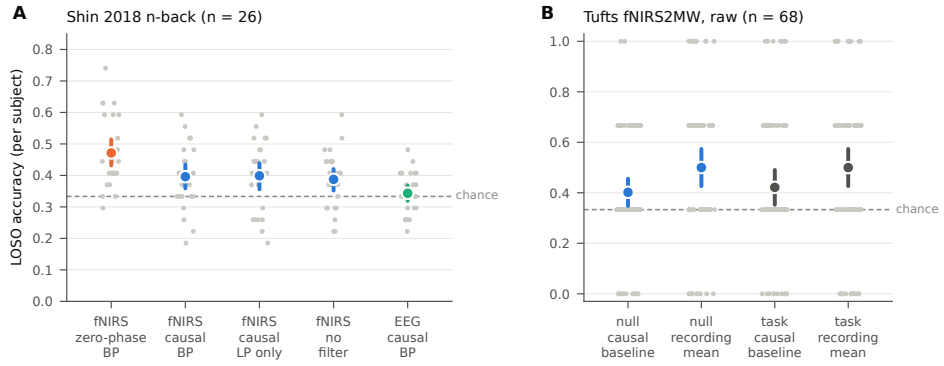


Figure 2: Null-window probe. Grey dots are participants. Coloured points and bars show the mean and bootstrap 95% CI. (A) Shin n-back: fNIRS under four filters and EEG; windows end 2 s before the instruction. BP, band-pass; LP, low-pass. (B) Tufts, raw blocks 14–16: null windows and task windows (20–24 s after onset) with a causal baseline or the recording mean. Per-subject accuracies are multiples of 1/3 because each participant contributes three blocks.

Table 2: Null-window probe, per-subject LOSO accuracy (mean), one-sided Wilcoxon test against chance.

Dataset (chance)	Setting	Accuracy	Excess	p
Shin fNIRS (33.3%)	zero-phase band-pass	47.2	+13.8	$< 10^{-4}$
	causal band-pass	39.6	+6.3	0.003
	causal low-pass only	39.9	+6.6	0.004
	no filter	38.8	+5.4	0.004
Shin EEG (33.3%)	causal band-pass	34.3	+1.0	0.24
Tufts fNIRS (33.3%)	causal baseline	40.2	+6.9	0.019
	recording mean	50.0	+16.7	5×10^{-5}
	no offset removal	31.9	−1.5	0.71
Shin WG fNIRS (50%)	per-subject order	51.9	+1.9	0.13

3.4. Controls

In the positive control, the probe stayed at chance with no injected drift (33.8%), detected half the observed drift (40.1%), and with the full observed drift reproduced the real-data accuracy (48.7% vs 48.0%; Fig. 3A). In Tufts' final blocks, a drift common to all participants is therefore enough to explain the leak. Because position and the preceding block are aliased there, these data cannot separate drift from carry-over.

In the negative control, the same participants, montage and causal pipeline gave chance-level fNIRS probes when order was randomised per participant (WG: 51.9%, $p = 0.13$; Fig. 3B), compared with +6.3 pp for n-back. The randomisation rule alone, without any brain data, would predict the current WG label with 62.2% ("opposite of previous") or 66.8% ("class with more trials remaining in the run") accuracy. Holding the previous label fixed, a small residual remained (balanced accuracy 52.7–54.2%), well below what the rule allows. Constrained randomisation that is shared by all participants is thus itself a weak shared channel.

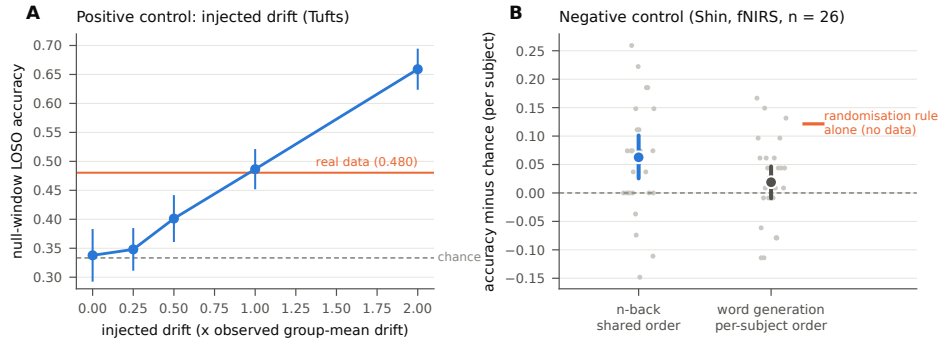


Figure 3: Controls. (A) Drift injection in Tufts: positions scrambled within participant, then k times the observed group-mean drift added (mean ± SD over 20 repetitions). (B) Same participants and causal pipeline: n-back (shared order) versus word generation (per-subject order), accuracy minus chance. The orange tick shows what the WG randomisation rule alone predicts.

3.5. Order information in task decoding (Tufts)

On the raw final blocks with recording-mean offset removal, a decoder trained only on rest-end windows, which contain no task data and precede the instruction, matched a task-trained decoder on task windows (0 vs 2-back: 70.4% vs 70.3%; three classes: 46.5% vs 47.1%). The task-trained decoder classified the rest before a block at 68.4% (Fig. 4A). With the causal

baseline the pattern held (rest-to-task 58.5%, $p = 3 \times 10^{-4}$; task-to-task 58.8%). A strong zero-phase high-pass (0.001 Hz) applied to the ∼600 s final-block recording abolished the transfer and lowered task decoding to 59.3%. This filter is far more aggressive than the official one, which was applied to the full ∼40 min session, so it does not represent the official preprocessing.

On the official data and protocol, a 0-vs-2-back logistic-regression decoder reached 64.2% over all blocks (67.8% on blocks 9–16). Its probability of "0-back" rose with session position at fixed true load (Fig. 4B). In the regression, the position effect accumulated over the session (+0.078) was as large as the 2- versus 0-back difference (−0.081). A preceding 3-back added +0.044 ($p = 7 \times 10^{-4}$). The only 0-back not preceded by a 3-back (position 1) was classified correctly in 44.9% of windows, against 64.9% for the other 0-backs ($p = 0.0006$). Removing the preceding-3-back shift lowered accuracy on blocks 9–16 from 67.8% to 63.7–64.0%, i.e. about a quarter of the above-chance margin. This is a coarse linear estimate from the same data.

The official DeepConvNet reached 62.0% (66.5% on blocks 9–16). Its output showed an even larger position effect (+0.0077 per block, $p = 5 \times 10^{-6}$; 0.115 over the session, 84% of the 2- vs 0-back difference) but no detectable preceding-3-back effect (+0.007, $p = 0.71$; Fig. 4C). Because Tufts' Latin square gives 0- and 2-back the same mean position, the position effect does not raise average accuracy in this protocol. Instead it produces position-dependent errors, for example 36% correct for the 0-back at position 8.

3.6. Reversed block pairs (Chen Stroop)

Without any pre-onset window, condition decoding from task windows was above chance for forward pairs (positions 1–2: 65.3%, $p = 0.001$; 3–4: 57.6%, $p = 0.035$) and at chance for the reversed pair (2–3: 47.2%). The paired difference between forward and reverse pairs was significant (61.5% vs 47.2%, $p = 0.011$, 15 of 18 participants; unfiltered: 60.4% vs 47.2%, $p = 0.012$; Fig. 5). Across participants, Stroop condition is decodable here only when it agrees with time in session. A decoder trained to separate early from late blocks, with condition balanced, also separated positions 2 and 3 (64.6%, $p = 0.01$–0.02).

3.7. Published methods

BenchNIRS. The published pipeline reproduced (LDA 39.2% versus 38.9% published). Shifting the markers 12 s earlier, so that features came from −12 to −2 s before onset, gave 37.1% under the published 5-fold scheme. Under LOSO this was 37.9% ($p = 0.015$), versus 39.2% ($p = 0.0014$) for the

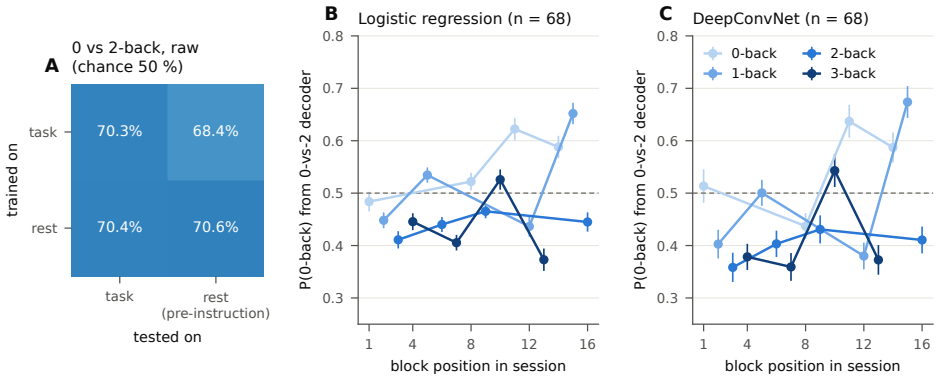


Figure 4: Tufts benchmark. (A) Transfer between task and rest-end windows, raw final blocks, 0 vs 2-back, recording-mean offset removal. (B, C) Probability of "0-back" from a 0-vs-2 decoder applied to all 16 blocks of held-out participants (official data and buckets), by position and true load (mean ± SEM over 68 participants). (B) Logistic regression; (C) the official DeepConvNet.



Figure 5: Chen Stroop: condition-decoding accuracy on forward pairs (neutral before incongruent) and on the reverse pair (incongruent before neutral). Lines connect participants; mean and bootstrap 95% CI shown.

published window, and the paired difference was not significant ($p = 0.74$; Fig. 6A). About 4.6 of the 5.8 pp above chance were available before the block began. A 16 s shift gave 35.5% ($p = 0.08$).

Con4m. The published result reproduced (F1 71.1% versus 71.28% published, one split). Re-concatenating blocks in per-participant random order did not change performance in any combination of training and test order (fixed→fixed 69.1 ± 1.4%, fixed→shuffled 69.4 ± 1.6%, shuffled→fixed 70.6 ± 1.6%, shuffled→shuffled 69.2 ± 1.4%; paired $p = 0.19$; Fig. 6B). Con4m uses the official task-only release, in which rest periods, the main carrier of order information here, have been removed.

Tufts DeepConvNet: see above.

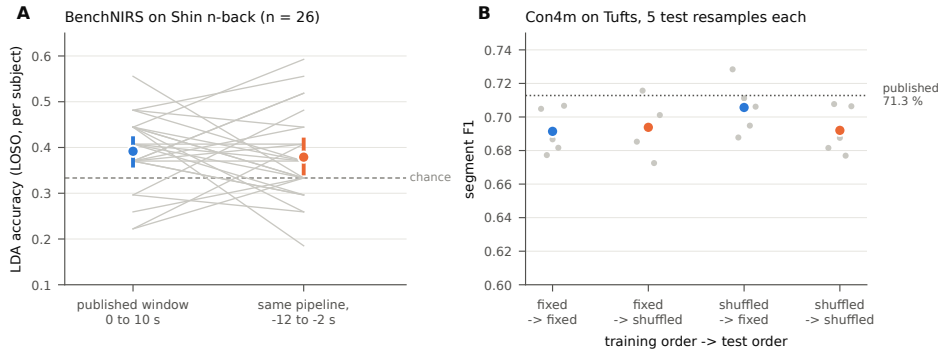


Figure 6: Re-running published methods. (A) BenchNIRS on Shin n-back: the published pipeline with the published window and with the same pipeline moved before block onset (LOSO, per participant). (B) Con4m on Tufts: segment F1 for every combination of training and test block order; grey dots are five test resamples, the dotted line is the published F1.

3.8. An uninformative dataset

In PhysioNet mental-fnirs ($n = 12$, one block per condition), no contrast, including 0- vs 2-back itself, was decodable across participants (43–57%). The dataset is too small to audit, and we list it in the census only.

4. Discussion

When every participant receives the same sequence, holding out a participant does not hold out the sequence. In three fNIRS datasets, information about the upcoming condition was present before the block began and before any cue. In the most conservative analyses it amounted to about six

Table 3: Published methods re-run under order-removing interventions.

Method	Dataset	Reproduced (published)	Effect of removing order information
BenchNIRS LDA [4]	Shin n-back	39.2% (38.9%)	37.9% from −12 to −2 s before onset; paired $p = 0.74$; ∼80% of the margin pre-onset
Tufts DeepConvNet [2]	Tufts	62.0% / 66.5%	position shifts output by 84% of the load effect; no preceding-block effect; mean accuracy not inflated
Con4m [13]	Tufts	F1 71.1% (71.28%)	no change with per-participant block order

to seven percentage points above chance. With conventional preprocessing it reached 12–17 points. In one benchmark, a decoder that never saw task data matched a task-trained decoder. A published benchmark result was largely available before onset.

Why subject-independent splits worked in earlier studies but not here. Li et al. and Ahmed et al. found chance-level accuracy when they analysed the OP1 image-EEG data across subjects [7, 8], and so did a "watermelon EEG" control [9]. Leakage across subjects requires temporal structure that is both time-locked to the session and consistent across people. Block-design workload tasks provide it: haemodynamic carry-over, recovery after demanding blocks, rest lengths that depend on load, and drift in fNIRS baselines. Rapid image presentation and inert objects do not. Consistent with this, the effect was clear in fNIRS and absent in EEG band power from the same windows.

The effect depends on the pipeline. Not all decoders exploit the shared order. Con4m was unaffected, which we attribute to the task-only release it consumes. DeepConvNet's output tracked session position strongly, but in Tufts' Latin square this did not inflate mean accuracy. The relevant variables are whether rest or pre-onset periods enter the analysis, whether windows are locked to block onset, the filter phase, and how between-subject offsets are removed. For this reason we propose a diagnostic rather than a correction factor.

Relation to prior work. Barbu et al. noted, for an fMRI study, that a fixed order may let cross-subject classifiers use preceding stimuli, and suggested randomising order or cross-validating by subject and run [11]. Within-subject temporal confounds in block designs [7, 8, 9], filter-

induced pre-onset decoding [10], pre-stimulus sanity checks [17], sequence-dependence explanations of pre-onset effects [18], design-variable probes [19], and shared-stimulus confounds in cross-subject benchmarks [20, 21, 22] are related. To our knowledge, none quantifies shared-order leakage under subject-independent evaluation in EEG/fNIRS benchmarks or offers a dedicated control for it.

Limitations. Effect sizes in conservative analyses are moderate. In single fixed sequences, position, preceding block and rest length are aliased, so the mechanism cannot be fully decomposed. Tufts' raw release limits the null-window probe to three blocks per participant. Con4m was run on one of twelve splits with one training seed. The DeepConvNet re-run deviated from the official grid for CPU time. The negative control shares participants and session with the positive dataset but uses a different task.

Recommendations. (1) Randomise condition order per participant, and do not apply one randomisation rule identically to all participants; if a rule is used, report the accuracy it alone permits. (2) Release per-participant event timing, including rest and cue periods, and check that the descriptor matches the data. (3) Accompany subject-independent results with a null-window probe ending before any cue, or a reversed-pair test where the design allows. (4) Avoid zero-phase filtering and per-block baselines that overlap the windows being compared.

5. Conclusions

Shared presentation order is common in public EEG/fNIRS datasets, and subject-independent evaluation does not neutralise it. A simple null-window probe, with the controls described here, can show whether a reported decoding result could have been obtained before the task began.

CRediT authorship contribution statement

Pinzhen He: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) to assist with literature searching, analysis code, figure generation and drafting of the manuscript text. After using this tool, the author reviewed and

edited the content as needed and takes full responsibility for the content of the publication.

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Declaration of competing interest

The author declares no competing interests.

Data availability

All datasets are public [1, 2, 3, 12]. Analysis code: `https://github.com/huixun160/shared-order-audit`.

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